Neural Network Field Guide

42 architectures and training schemes, from the perceptron (1958) to diffusion LLMs (2025).

Each diagram draws a network in miniature, from input at the bottom to output at the top. In most panels, a circle is one neuron; in sequence models, a column is a token and a circle its whole vector at one layer. A square is larger: an image patch, a matrix-valued state, a whole network (in gray). Color and mark both show a cell's role, so diagrams read without color. Faded parts play little or no role in the step shown. Parts repeated over positions, windows, copies, or passes share weights; layers, experts, encoders, and decoders drawn alike do not. Sections group by a defining trait: wiring, dynamics, training, cost, or purpose. Years and training methods follow landmark papers, not always the first; many can be trained other ways.

Input
- Input unit
- Image patch
- Noisy input
- Random noise
- Visible unit (in and out)
- Masked token

Hidden
- Hidden unit
- Probabilistic hidden
- Capsule (vector output)
- Router

State
- Recurrent unit
- Memory cell
- Matrix-valued state

Mixing
- Attention
- Convolution or pooling

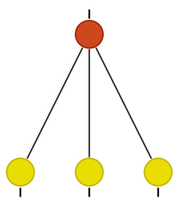
Output
- Output unit
- Reconstruction of input
- Velocity

Network
- A whole network

Links
- Connection
- Skip connection
- Data passed on or fed back
- Gate, acting at the dot
- Cross-attention, comparison
- Learnable function

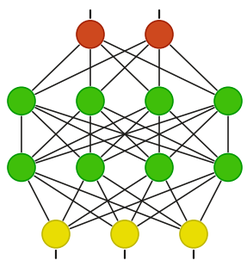
FEED-FORWARD

Perceptron



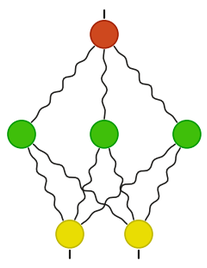
Weighted sum, then a threshold
1958 · supervised

MLP



Stacked dense layers
1986 · supervised

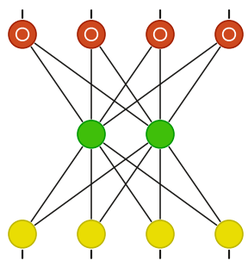
KAN



Learnable functions on edges
2024 · supervised

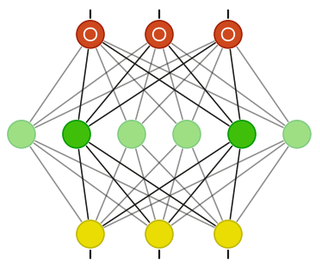
AUTOENCODERS

Autoencoder



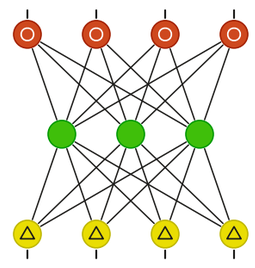
Encode, then reconstruct
1986 · reconstruct

Sparse autoencoder



Few active units per input
2006 · reconstruct

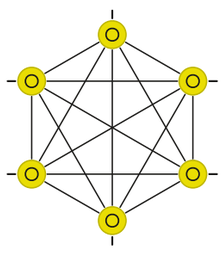
Denoising autoencoder



Clean up a noisy input
2008 · denoise

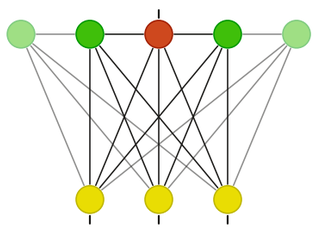
ENERGY-BASED AND SELF-ORGANIZING

Hopfield network



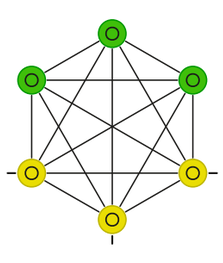
Settles toward a stored pattern
1982 · hebbian

Kohonen map



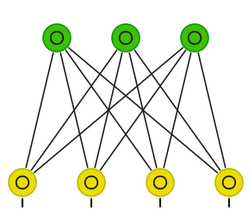
Winner and neighbors move
1982 · competitive

Boltzmann machine



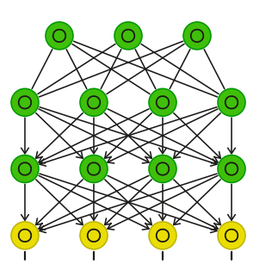
Stochastic units, fully connected
1985 · likelihood

RBM



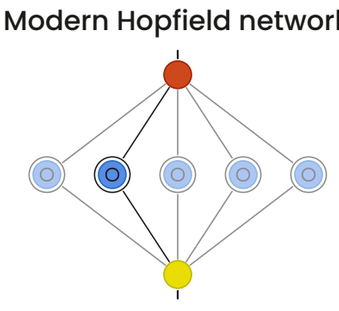
No links within a layer
1986 · likelihood

Deep belief network



Stacked RBMs, trained greedily
2006 · likelihood

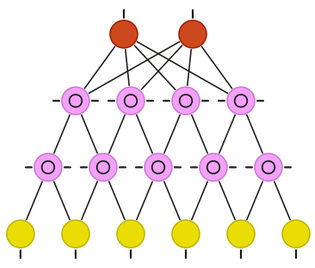
Modern Hopfield network



Attention as memory lookup
2020 · supervised

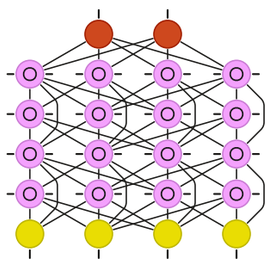
CONVOLUTIONAL

CNN



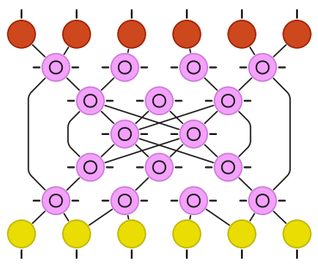
Local filters, shared weights
1989 · supervised

ResNet



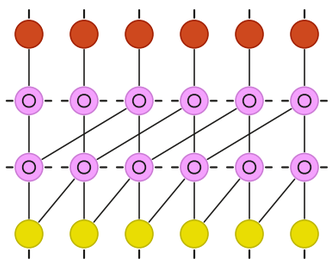
Blocks add to a skip path
2015 · supervised

U-Net



Shrink, expand, skip across
2015 · supervised

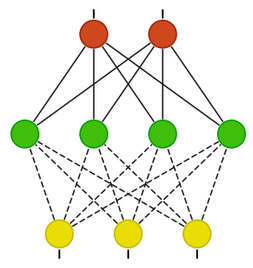
WaveNet



Dilated causal convolutions
2016 · next token

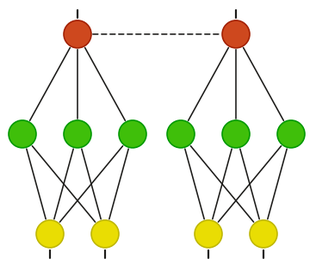
SIMILARITY

Radial basis network



Distance to stored centers
1988 · supervised

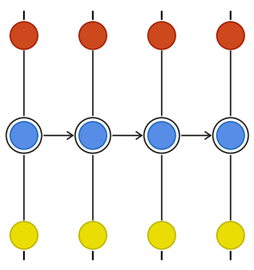
Siamese network



Twin towers, shared weights
1993 · contrastive

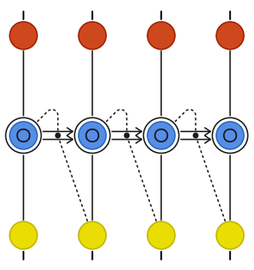
RECURRENT AND MEMORY

RNN



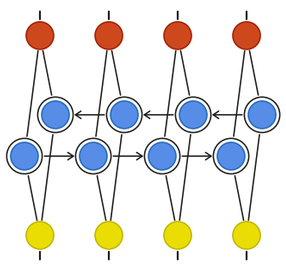
Hidden state passed along
1990 · next token

LSTM



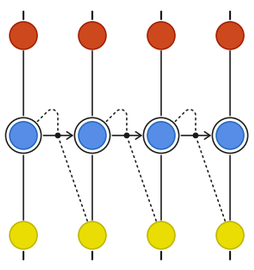
Two states; forget gate from 1999
1997 · next token

Bidirectional RNN



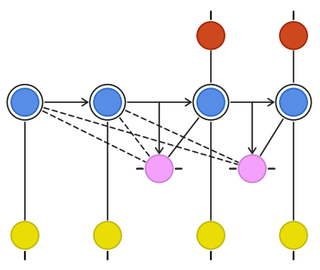
Reads the sequence both ways
1997 · supervised

GRU



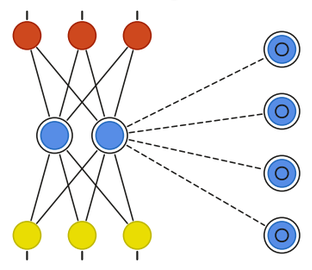
One state, two gates
2014 · next token

Seq2seq with attention



Learns to align and translate
2014 · next token

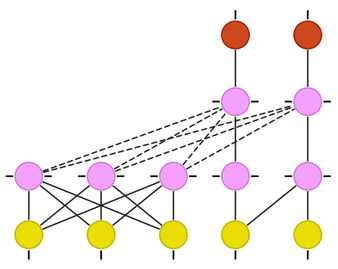
Neural Turing machine



Controller reads, writes memory
2014 · supervised

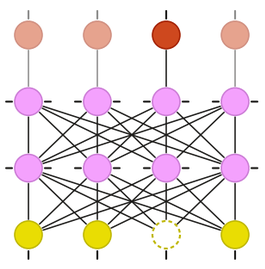
ATTENTION

Transformer



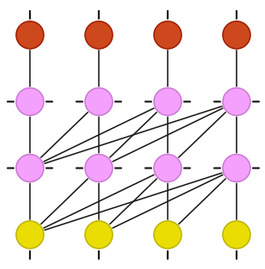
Encoder, decoder, cross-attention
2017 · next token

BERT



Every token sees every token
2018 · masked token

GPT-style LLM



Sees itself and earlier tokens
2018 · next token

ViT

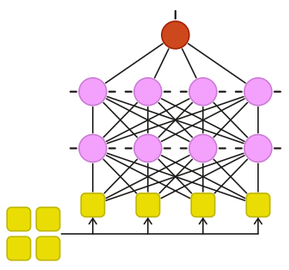
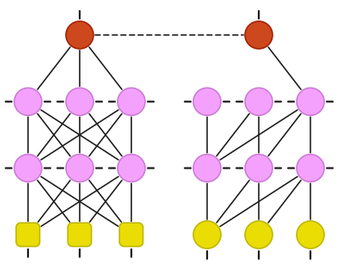


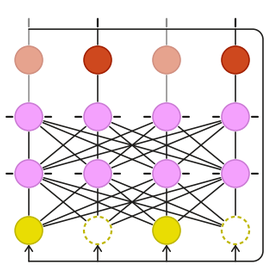
Image patches become tokens
2020 · supervised

CLIP



Match images to captions
2021 · contrastive

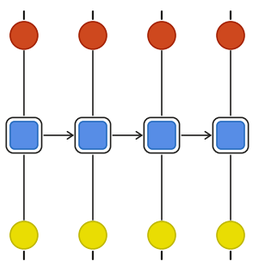
Masked diffusion LLM



Unmask in parallel, step by step
2025 · masked token

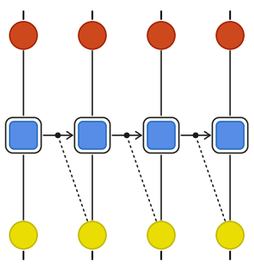
LINEAR-TIME MODELS AND HYBRIDS

Linear attention



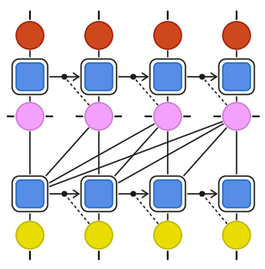
Softmax-free attention as an RNN
2020 · next token

Mamba



Selective state, linear time
2023 · next token

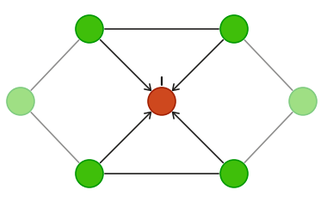
Mamba–attention hybrid



Mostly recurrent, some attention
2024 · next token

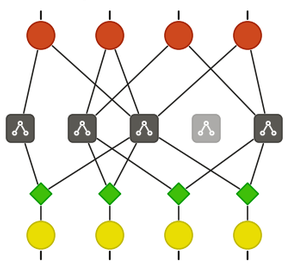
ROUTING AND GRAPHS

Graph neural network



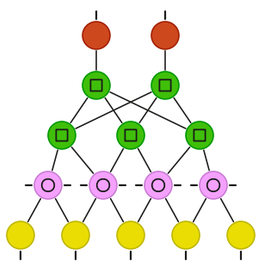
Nodes pool neighbor messages
2005 · supervised

Sparse MoE



Each token → its top-*k* experts
2017 · next token

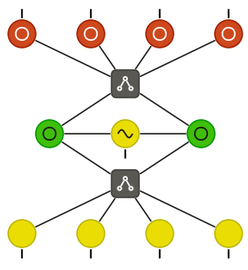
Capsule network



Vectors routed by agreement
2017 · supervised

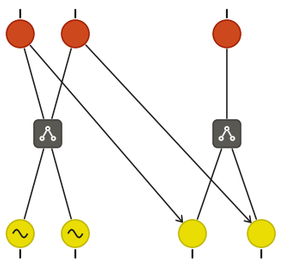
GENERATIVE FRAMEWORKS

Variational autoencoder



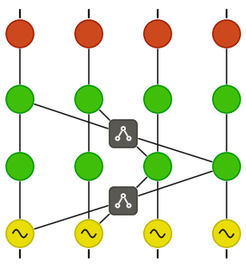
Sample a code, then decode
2013 · variational

GAN



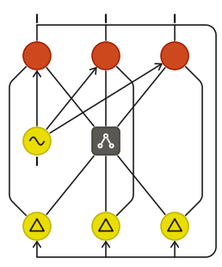
Generator vs. discriminator
2014 · adversarial

Normalizing flow



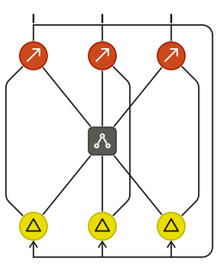
Invertible map from noise to data
2014 · likelihood

Diffusion model



One denoiser, run *T* times
2015 · variational

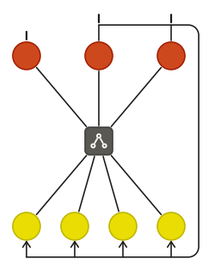
Flow matching



Learns a velocity, noise to data
2022 · flow matching

REINFORCEMENT

Policy gradient (PPO)



More of what earned reward
2017 · reward

Inspired by The Neural Network Zoo by Fjodor van Veen (Asimov Institute, 2016; updated with Stefan Leijnen, 2019). www.asimovinstitute.org/neural-network-zoo

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